Large-scale traffic flow reconstruction from heterogeneous mobility sensors

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Abstract

Mobile-phone data, traffic counters and GPS traces give complementary and incomplete views of road traffic. We combine them to reconstruct hourly flows on the regional road network of Emilia-Romagna, Italy. Mobile-phone origin–destination activity drives an agent-based simulation on an OpenStreetMap graph; inductive loop detectors correct the simulated counts locally, and the correction is propagated to neighbouring links with simulated turning shares. Viasat GPS probes are kept out of the reconstruction and used for validation after a link-specific penetration rescaling. On all usable matched links, 848 in a February 2020 week and 845 in a June–July 2021 week, the median absolute error of the hourly traffic shares is 1.34 and 1.49 percentage points, 32% and 36% of the mean hourly share, and 27% on the links with penetration near 1%. The simulated peaks precede the observed ones by about one hour. The reconstruction improves only slightly on the hourly profile of the injected demand (median error 1.40 and 1.52 points) and is less accurate than the mean profile of the regional loop detectors (1.15 and 1.21 points). In a leave-one-out test on 149 detectors, assimilating the others lowers the error on the held-out detector from 59.8% to 56.7% of its observed volume. Used as an exposure denominator for the 2022–2024 accident tables, the reconstructed field gives a relative risk that peaks around midnight and an accident share that scales across hours as $E^{0.72}$ (95% interval 0.58–0.88). The workflow runs as a webserver that delivers the assimilated link flows.

Keywords: traffic reconstruction, mobile-phone data, traffic counters, GPS telemetry, agent-based model, data fusion

1 Introduction

Passive mobility data have changed the scale at which urban and regional transport systems can be observed. This shift is part of a broader movement toward urban computing, smart-city analytics and digital representations of urban systems [1–5]. Mobile-phone records show regular spatial and temporal patterns in human movement [6, 7] and have been used to estimate origin–destination (OD) matrices for transport analysis [8–10]. More generally, large geolocated data sets can support travel-demand estimation and road-use analysis when they are combined with network information and independent validation [11–13]. **[TODO: cite earlier Bologna-group papers on Emilia-Romagna mobility; none are in the repository]** These data streams do not, however, remove the central inverse problem of traffic reconstruction: demand is observed at population or areal scale, whereas congestion, emissions and infrastructure assessment depend on flows on individual road links [14–16].

A link-level traffic field is also an exposure denominator for road-safety analysis. Crash-frequency and before–after studies use traffic volume or vehicle distance travelled to interpret accident counts, since more accidents on a road or in an hour may simply reflect more vehicles [17–20]. Speed, traffic regime and road environment modulate this relation [21, 22]. A reconstructed field can normalize aggregate accident data in time and weight the simulated traffic of each link by the hours in which it occurs.

Each observational layer has a different failure mode. Mobile-phone OD data provide broad coverage but do not identify the exact road sequence followed by a trip, and trip-distribution models are known to be sensitive to spatial scale and distance-decay assumptions [23–26]. Fixed traffic counters measure vehicle passages directly, yet only at instrumented locations; using them to correct an OD matrix is a classical inverse problem with many more unknowns than counts [27]. Floating-car or GPS telemetry is spatially richer, but it measures the vehicles present in the provider's fleet and therefore requires a road-dependent penetration normalization before comparison with total traffic [28–31]. Treating these layers as interchangeable counts can produce non-physical aggregate validations, for example by summing flows on roads that do not represent the same traffic stream. The same issue appears more generally in heterogeneous urban data, where missing values, sampling mechanisms and data governance shape what can be inferred [32–34].

We build a link-level reconstruction workflow in which each data source keeps its own role. Mobile-phone OD activity generates time-dependent demand. A directed road graph built from OpenStreetMap [35, 36] supports shortest-path routing [37] in the traffic-assignment tradition [14, 15]. Inductive loop detectors of the regional traffic monitoring system (MTS), which we call coils below, correct the simulated link counts where they are installed. Viasat GPS telemetry stays outside reconstruction and assimilation and is used only for validation, after a link-specific penetration coefficient removes the size of the probe fleet. We validate the hourly profiles on all usable matched links of a 2020 and a 2021 week, compare them with two baseline profiles, test the assimilation by leaving out one coil at a time, and use Ward clustering [38, 39] to choose example links with different daily profiles. The same field is served by a webserver to analysts who do not run the modelling chain [2, 5], and we use it as an exposure layer for aggregate accident data.

Section 2 presents the data layers, the reconstruction, the accident exposure analysis, the validation against Viasat and the webserver. Section 3 describes data, model, assimilation, risk indicators and validation procedure. Section 4 discusses the results and their limitations, and Section 5 concludes. Tables and further details are in the Supplementary Material.

2 Results and validation

Figure 1 summarizes the reconstruction. Mobile-phone data define the demand, the road graph defines the admissible movements, coils constrain local volumes through assimilation, and GPS traces are held out for validation. Information from the fixed counters spreads along the simulated turning structure, so it follows the direction of traffic.

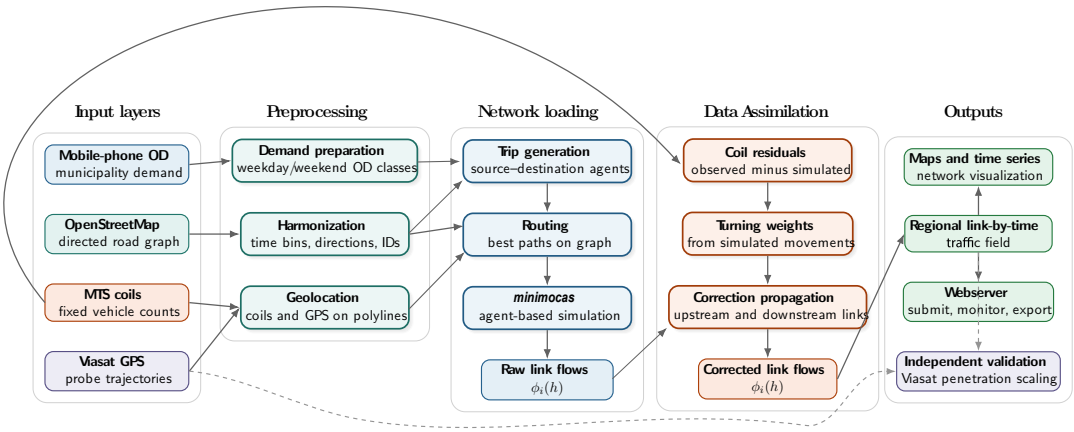


Fig. 1 End-to-end reconstruction workflow. Mobile-phone OD activity defines regional demand, OpenStreetMap provides the directed road graph, MTS coils are geolocated on road polylines and Viasat GPS traces are kept for independent validation of the simulation framework. Demand is injected into the `minimocas` agent-based simulator, producing raw link flows. Coil residuals are then used for local data assimilation of the simulated counts and propagated through adjacent directed links using simulated turning weights. The final product is a data-assimilated link-by-time traffic field that can be mapped, validated against held-out GPS traces and exposed through the webserver.

2.1 Observation layers and mobile-phone demand

The reconstruction is carried out on the Emilia-Romagna regional road network, whose main structure is an east–west corridor connected to dense urban areas, coastal roads and Apennine links (Fig. 2a). The mobile-phone layer is aggregated at municipality level and converted into OD activity. Weekday activity in January 2020 is strongly concentrated along the regional urban corridor, especially around Bologna and the main provincial centres (Fig. 2b). The model has to redistribute this demand onto individual road segments [8, 11, 15].

The MTS coils provide local vehicle counts on selected roads (Fig. 2c). Coil 282 is used as an illustrative fixed counter because it has a complete directional time series for the selected day and an unambiguous geographical record. Its 15-min profile has low night traffic, a morning peak at 07:00 in direction 1 and an evening peak at 17:00 in direction 0 (Fig. 2d). Sparse fixed detectors and broader but sampled mobile traces are the typical inputs of traffic-state reconstruction [28, 31].

The demand layer also contains temporal and categorical information. Mean weekday activity has a sharper morning peak and a stronger late-afternoon component than weekend activity (Fig. 3a). The trip-generation classes indicate whether both endpoints are observed in the OD data or whether one or both endpoints are sampled from the modelled area (see Methods, Section 3.1). Figure 3b,c shows the mobility components used to decompose agent behaviour from the OD data. In both weekdays and weekends, the source–destination component dominates the reconstructed demand, while random-endpoint components represent residual mobility that cannot be directly linked to two observed municipalities. They spread weakly observed OD mass over the network, so that it does not pile up on a few high-count municipality pairs. Ranking municipalities by activity confirms that a small number of urban centres drive a large fraction of the demand (Fig. 3d), consistent with empirical evidence that human mobility is highly heterogeneous across places and distances [6, 23, 24].

2.2 Network reconstruction and local counter control

The simulation converts the OD demand into directional link flows. For each hour, generated agents are placed on the directed road graph, routed toward their destination and counted on every traversed polyline. The

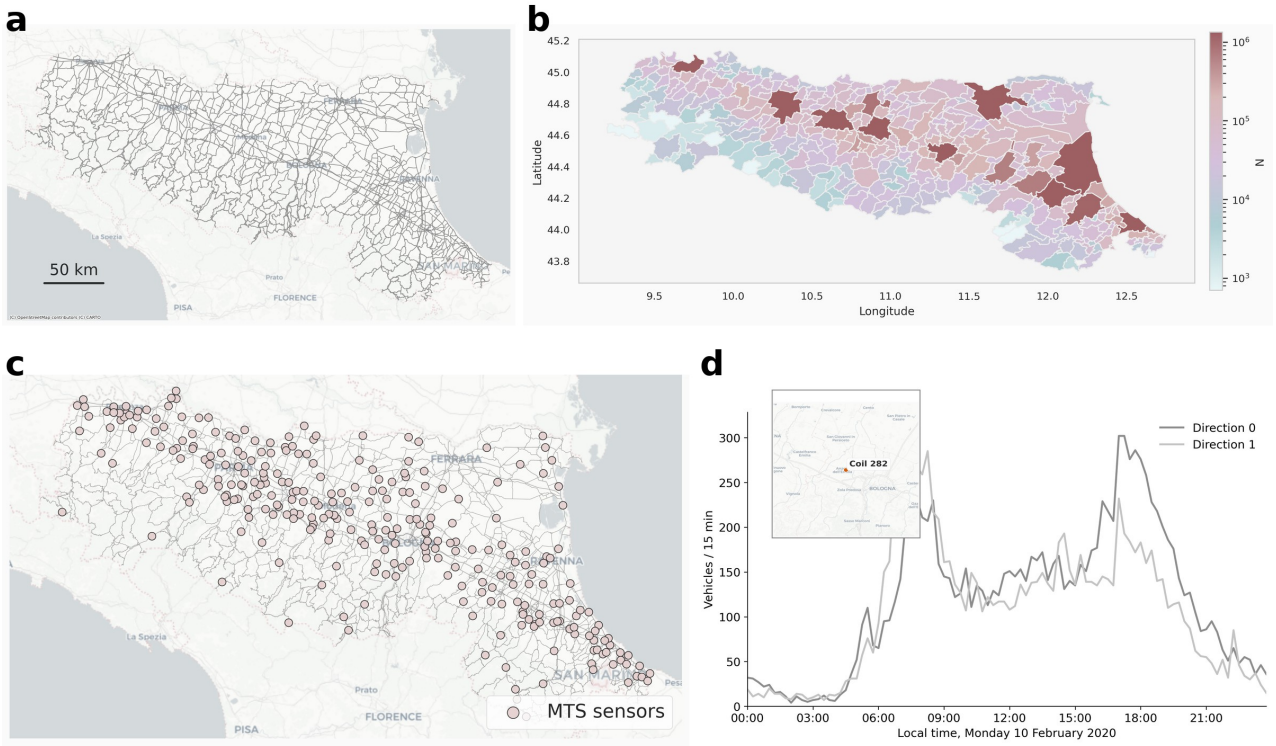


Fig. 2 Study domain and observational inputs. (a) Regional road cartography used by the model on a CartoDB Positron basemap; the scale bar corresponds to 50 km. **(b)** Weekday mobile-phone OD activity over Emilia-Romagna municipalities in January 2020; colour encodes activity N on a logarithmic scale. **(c)** Locations of the MTS inductive loop detectors (coils) over the regional road graph. **(d)** Location of coil 282 (inset) and complete 15-min directional vehicle counts on Monday 10 February 2020, local time.

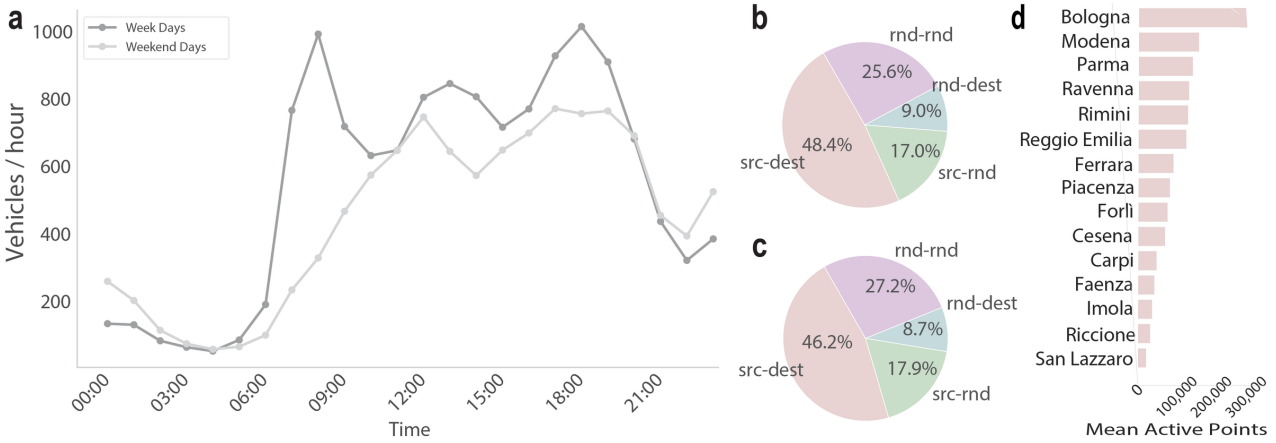


Fig. 3 Mobile-phone demand used to drive the reconstruction. (a) Mean hourly OD activity profiles for weekdays and weekend days in January 2020. **(b)** Weekday and **(c)** weekend shares of the four generation classes: source–destination (`src-dest`), source–random (`src-rnd`), random–destination (`rnd-dest`) and random–random (`rnd-rnd`). **(d)** Mean mobile activity of the 15 most active municipalities.

output is a link-by-time matrix $\phi_i(h)$, where i is a road polyline and h is the hour. On Monday 10 February 2020, a working day before the COVID-19 mobility restrictions, the raw simulated flow on the polyline of coil 282 follows the daily structure of the counter (hourly correlation 0.95) with 67% of its daily volume, 14,804 against 21,935 vehicles (Fig. 4a). After assimilation the link carries the coil series by construction; the one-hour offset between the two curves comes from the UTC clock of the coil table (Section 4.1). The map of simulated daily flows highlights the high-volume regional corridors while retaining non-zero information on the secondary network (Fig. 4b). This microscopic-to-link aggregation follows the logic of agent-based traffic simulators, where individual vehicle trajectories are used to recover network-level flow fields [40–42].

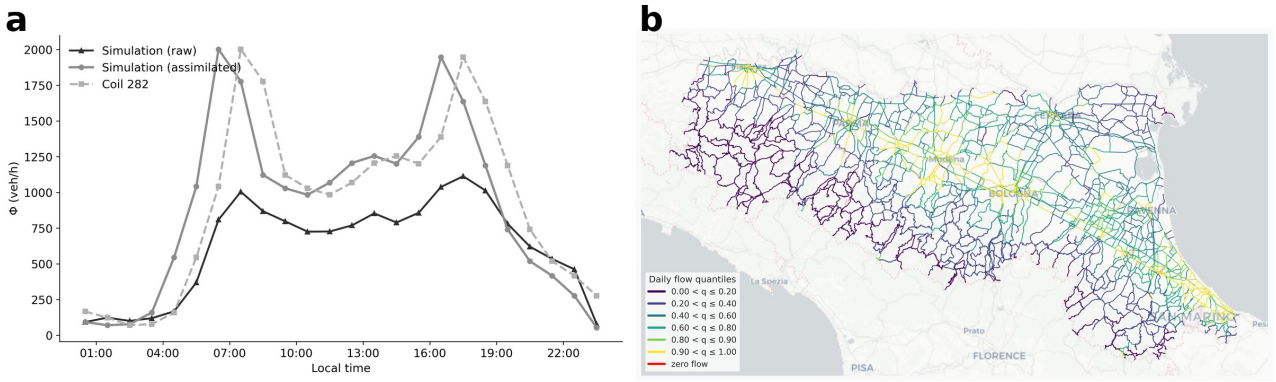


Fig. 4 Link-level reconstruction output. (a) Raw and assimilated simulated flow on the polyline of coil 282 and hourly coil counts on Monday 10 February 2020, local time; Φ denotes vehicle flow. **(b)** Spatial distribution of simulated daily link flows over Emilia-Romagna, expressed through flow quantile classes; yellow segments carry the highest reconstructed flows and purple segments the lowest.

The assimilation corrects the coil links themselves; how far the correction reaches is measured by a leave-one-out test on 10–17 February 2020. For each of the 149 coils with complete data on at least one day (610 coil-days), we remove the coil from the assimilation and from the set of polylines that stop the propagation, rerun the assimilation with the other coils, and compare the hourly counts on its link with the observation (Supplementary Section S4). Summed over all coil-days, the absolute error on the held-out link is 59.8% of the observed volume for the raw simulation and 56.7% after assimilation; the median over coils goes from 46.2% to 44.3%. The propagation from neighbouring coils reaches 51% of the held-out links, and the error decreases on 39% of the coils and increases on 13%. On the Viasat links the median profile error defined in Section 2.5 is 1.33 pp before and 1.34 pp after assimilation (Table 2). The assimilation therefore anchors the field at the instrumented sections and has a small effect a few links away from them.

2.3 Accident exposure

The reconstructed field can serve as an exposure denominator for aggregate accident statistics. The regional accident tables for 2022–2024 (16,679, 16,808 and 16,757 accidents) give counts by hour of the day and by road type, with no location. Let $A(h)$ be the share of accidents in hour h and $E(h)$ the share of simulated vehicles on the whole network in the same hour (Fig. 5a). We define the relative risk

$$R(h) = \frac{A(h)/E(h)}{\frac{1}{24}\sum_{h'} A(h')/E(h')}, \tag{1}$$

which equals one on average over the day. With weekday exposure, R is 3.4 at 00:00 and 2.4 at 01:00, stays between 1.0 and 1.4 from 03:00 to 07:00, and is below 0.6 after 19:00 (Fig. 5b). The night values rest on the simulated night exposure, the least reliable part of the simulated profile (Section 4.1). Ward clustering of the hourly risk and exposure values separates three regimes: night (21:00–06:00), morning (07:00–09:00) and day (10:00–20:00) (Fig. 5c).

A least-squares fit of $\log A$ against $\log E$ on the 24 hourly points gives $A \sim E^{\alpha}$ with $\alpha = 0.72$, 95% bootstrap confidence interval 0.58–0.88 (10,000 resamples of the 24 hours), and $R^2 = 0.85$ (Fig. 5d). Across hours, the accident share grows less than proportionally with the simulated exposure.

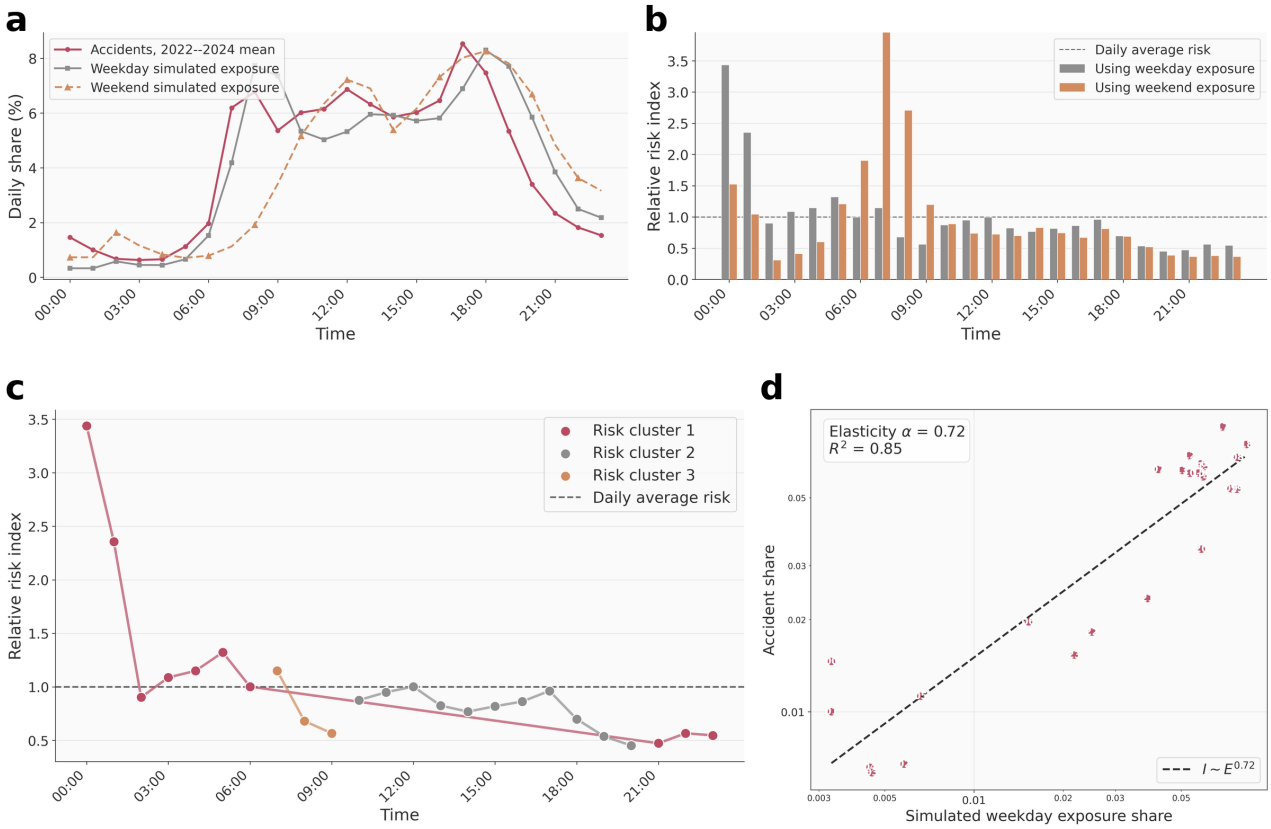


Fig. 5 Accidents and simulated exposure by hour. (a) Hourly accident share (mean of 2022–2024) and simulated weekday and weekend exposure shares. **(b)** Relative risk $R(h)$, Eq. (1), with weekday (grey) and weekend (orange) exposure; the dashed line marks $R = 1$. **(c)** Three hourly regimes from Ward clustering of risk and exposure values. **(d)** Accident share against simulated weekday exposure share for the 24 hours (log–log axes) and fit $A \sim E^{\alpha}$, $\alpha = 0.72$ (95% bootstrap interval 0.58–0.88), $R^2 = 0.85$.

$R(h)$ is a regional profile and is the same for every link. The hazard-weighted exposure H_i of a link (Eq. (12), Fig. 6a) is therefore its simulated volume weighted by the hourly composition of its traffic, and the rank difference S_i (Eq. (15), Fig. 6e) picks out links that carry a larger share of their traffic in high-R hours. The risk centrality C_i (Fig. 6c,d) and the weekday–weekend contrast D_i (Fig. 6b) follow the same logic. These maps show where simulated traffic coincides with hours of high regional risk; they carry no information on where accidents occur. Supplementary Fig. S3 gives the yearly totals, road types and seasonal shares of the accident data.

2.4 Independent Viasat validation

Viasat GPS observations give an independent test of the hourly profiles produced by the reconstruction; they enter neither the demand nor the assimilation. Matched Viasat links cover only part of the regional cartography, mostly in the Forlì–Cesena area (Fig. 7a,d). For each matched polyline i we estimate the penetration coefficient

$$\eta_i = \frac{\sum_d \sum_h \psi_i(d,h)}{\sum_d \sum_h \phi_i(d,h)}, \tag{2}$$

where $\psi_i(d,h)$ is the number of Viasat probes on day d and hour h, $\phi_i(d,h)$ is the simulated flow, and both sums run over the same days of the validation interval. The penetration-normalized GPS profile is

$$\widetilde{\psi}_i(h) = \frac{1}{\eta_i} \sum_d \psi_i(d,h). \tag{3}$$

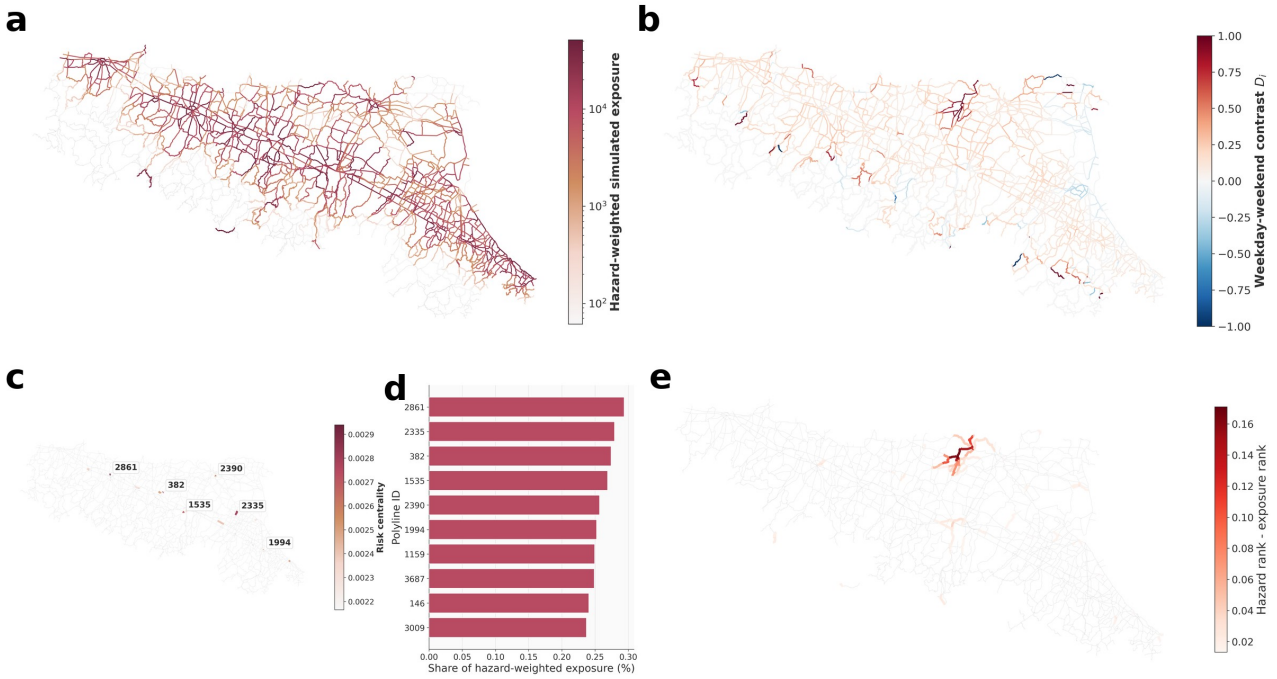


Fig. 6 Simulated weekday exposure weighted by the regional risk profile. (a) Hazard-weighted exposure H_i, Eq. (12). **(b)** Weekday–weekend contrast D_i, Eq. (13); positive values mark links with larger weekday H_i. **(c)** Location of the links with the highest risk centrality C_i, Eq. (14), and **(d)** their share of the total hazard-weighted exposure. **(e)** Links in the top 5% of the rank difference S_i, Eq. (15).

The rescaling sets the GPS volume of each road equal to the simulated one and leaves the hourly shape unchanged, so the comparison tests when traffic occurs on each link and says nothing about the size of the Viasat fleet. A link-specific factor is needed because probe coverage varies strongly across the network [28, 30, 31].

Validation compares the hourly shares $p_i(h) = \phi_i(h)/\sum_{h'} \phi_i(h')$ and $q_i(h) = \widetilde{\psi}_i(h)/\sum_{h'} \widetilde{\psi}_i(h')$, where $\phi_i(h) = \sum_d \phi_i(d,h)$; the shares do not depend on η_i. The profile error of link i is the mean absolute error over the 24 hours, in percentage points (pp),

$$\mathrm{MAE}_i = \frac{100}{24} \sum_{h=0}^{23} \big| p_i(h) - q_i(h) \big|, \tag{4}$$

and we also report it relative to the mean hourly share $100/24 \simeq 4.17$ pp,

$$\varepsilon_i = \frac{\mathrm{MAE}_i}{100/24} = \sum_{h=0}^{23} \big| p_i(h) - q_i(h) \big|, \tag{5}$$

which is the L^1 distance between the two daily distributions: $\varepsilon_i = 0$ for identical profiles and $\varepsilon_i = 2$ for profiles with disjoint support.

The penetration spans four decades (Fig. 7b,e). In 2020 the median η_i is 0.50%; the classes $\eta \sim 10^{-2}$ and $\eta \sim 10^{-3}$ contain 60.6% and 32.4% of the 848 usable links, and only 17 links (2.0%) reach $\eta \sim 10^{-1}$. In 2021 the median is 0.44% and the two main classes contain 61.3% and 34.2% of 845 links. The classes are interwoven in space (Fig. 7c,f): penetration is a property of individual roads more than of an area.

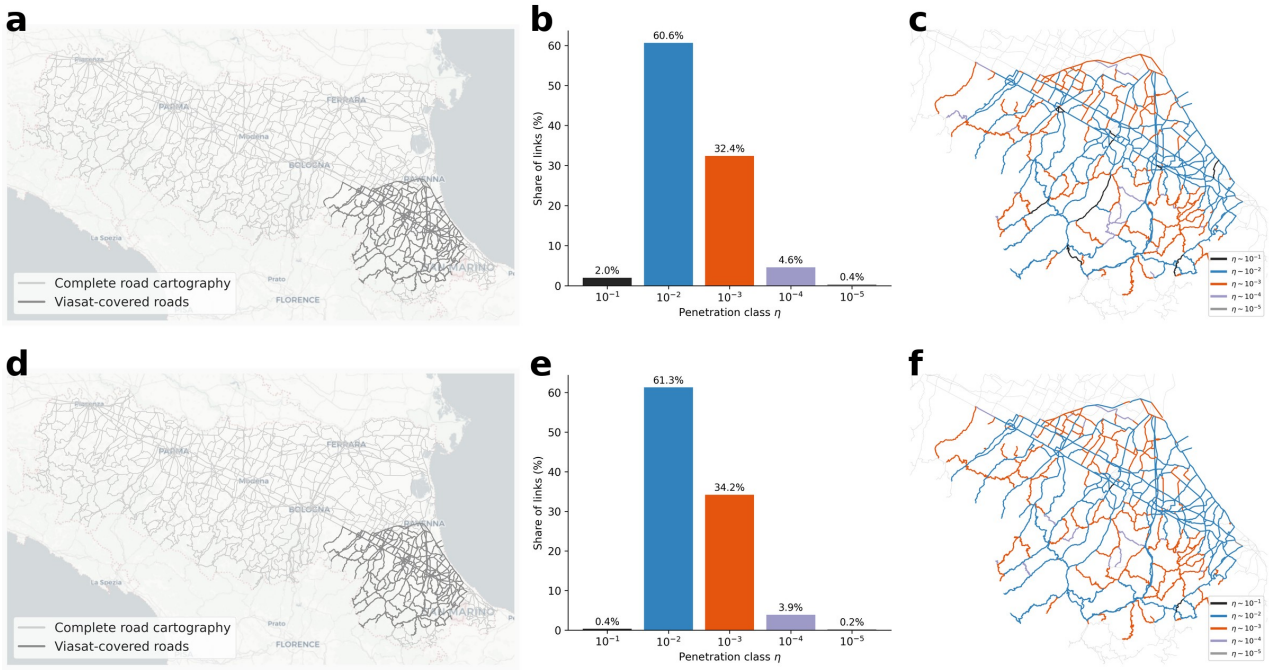


Fig. 7 Viasat coverage and link penetration. Top row: validation interval 10–17 February 2020. Bottom row: validation interval 28 June–4 July 2021. **(a,d)** Matched Viasat links (dark grey) over the complete regional road graph (light grey). **(b,e)** Share of usable links in each penetration class, η_i rounded to the nearest power of ten. **(c,f)** Geographical distribution of the same classes in the Forlì–Cesena validation area.

2.5 Profile errors on all matched links

We compute MAE_i on every matched link with non-zero simulated and GPS totals, 848 of the 930 matched links in 2020 and 845 of 927 in 2021. Simulation and GPS cover the same days: eight in 2020 (six weekdays and the weekend of 15–16 February) and seven in 2021 (five weekdays and the weekend of 3–4 July), with weekend days simulated from the weekend OD matrix. Table 1 and Fig. 8 summarize the distributions.

Table 1 Profile error of the assimilated simulation against penetration-normalized Viasat profiles on all usable matched links, by campaign and penetration class η. MAE in percentage points (Eq. (4)); ε is the median relative error (Eq. (5)); r is the median Pearson correlation of the hourly shares. Classes with fewer than ten links (three links with $\eta \sim 10^{-5}$ in 2020, two in 2021) are included in the totals only.

Campaign	Class	Links	Median MAE	IQR	10th–90th pct.	ε [%]	r
2020	all	848	1.34	1.05–2.01	0.87–3.28	32	0.82
	10^{-1}	17	0.94	0.91–0.97	0.79–1.12	23	0.91
	10^{-2}	514	1.13	0.96–1.34	0.83–1.61	27	0.87
	10^{-3}	275	2.13	1.66–2.82	1.43–3.65	51	0.66
	10^{-4}	39	5.11	4.63–6.77	3.77–7.50	123	0.36
2021	all	845	1.49	1.15–2.12	0.98–3.48	36	0.76
	10^{-1}	3	1.13	–	–	27	0.85
	10^{-2}	518	1.24	1.03–1.51	0.89–1.86	30	0.83
	10^{-3}	289	2.18	1.73–3.06	1.43–3.89	52	0.64
	10^{-4}	33	5.75	4.86–6.82	4.67–7.64	138	0.27

The median MAE is 1.34 pp in 2020 and 1.49 pp in 2021, i.e. a relative error ε of 32% and 36% of the mean hourly share. The error depends strongly on penetration. On the 514 links with $\eta \sim 10^{-2}$ the 2020 median is 1.13 pp ($\varepsilon = 27\%$), on the 275 links with $\eta \sim 10^{-3}$ it is 2.13 pp ($\varepsilon = 51\%$), and on the 39 links with $\eta \sim 10^{-4}$ the GPS profile is dominated by counting noise ($\varepsilon = 123\%$). The 2021 values are similar (Table 1).

Part of the error is a systematic lead of the simulation. Summed over all usable links, the simulated profile peaks at 17:00 and the Viasat profile at 18:00 in both campaigns, and the MAE of the two aggregate profiles is

0.69 pp (2020) and 0.82 pp (2021). Shifting the simulated profiles forward by one hour lowers the median link MAE to 1.29 and 1.34 pp. We do not apply this shift, since the simulator output and the Viasat tables both label each hour by its start; the possible origins of the lead are discussed in Section 4.1.

To judge these values we apply two fixed profiles to every link and compute the same error (Table 2). The first is the hourly profile of the OD demand injected in the runs, summed over the same days; it measures what the network loading adds to the timing of the demand. The second is the mean normalized hourly profile of the MTS coils: the 82 coil directions with complete data on all eight days in 2020 and, since no observed coil table covers June–July 2021, the summer weekday averages of 471 coil directions from the 2022 archive. The reconstruction has a lower error than the OD profile on 74% of the links in 2020 and 76% in 2021, but the gain is small, from 1.40 to 1.34 pp and from 1.52 to 1.49 pp in median. The MTS mean profile does better than the reconstruction: its median MAE is 1.15 pp in 2020 and 1.21 pp in 2021, and its error is lower on 91% and 87% of the links. On the $\eta \sim 10^{-2}$ class of 2020 the medians are 0.82 pp for the MTS profile and 1.13 pp for the reconstruction. A generic counter profile therefore describes the timing of traffic on the Viasat links better than the simulation, which takes its timing from the OD demand and anticipates the observed peaks by about one hour. What the reconstruction adds is a volume for every directed link and its spatial distribution, which no single profile provides; its hourly shape on unobserved links is its weakest component.

Table 2 Median and interquartile range of the link MAE (pp) for the reconstruction and for two profiles applied identically to every link: the hourly profile of the injected OD demand and the mean normalized profile of the MTS coils (2020: the 82 coil directions with complete data on all eight validation days; 2021: summer weekday averages of 471 coil directions from the 2022 MTS archive). The last column gives the percentage of links on which the assimilated simulation has a lower MAE than the baseline.

Campaign	Profile	Median MAE	IQR	Simulation better [%]
2020	Simulation (assimilated)	1.34	1.05–2.01	–
	Simulation (raw)	1.33	1.02–1.99	–
	Simulation shifted by +1 h	1.29	0.92–1.97	–
	OD demand profile	1.40	1.10–2.03	74
	MTS mean profile	1.15	0.74–1.85	9
2021	Simulation (assimilated)	1.49	1.15–2.12	–
	Simulation (raw)	1.39	1.09–2.05	–
	Simulation shifted by +1 h	1.34	1.04–2.01	–
	OD demand profile	1.52	1.21–2.18	76
	MTS mean profile	1.21	0.84–1.93	13

2.6 Temporal subclusters and representative roads

To show individual comparisons we take the 487 links of the upper class of a three-class k-means partition of $\log_{10} \eta_i$, which corresponds to $\eta_i \geq 0.40\%$ in 2020. Their simulated profiles are normalized to hourly shares and clustered with Ward linkage and Euclidean distance [38, 39]; normalizing removes the volume of each road, so the clusters group links by the shape of their daily profile. Cutting the dendrogram into three clusters (Ward distance 0.15, Fig. 9a) gives subclusters of 51, 78 and 358 links. Subcluster 1 has the sharpest morning peak (7.1% of daily traffic at 07:00) and the fastest evening decline, subcluster 3 the flattest morning and the largest evening tail, and subcluster 2 lies in between (Supplementary Fig. S2). The three subclusters are spread over the whole validation area (Fig. 9b).

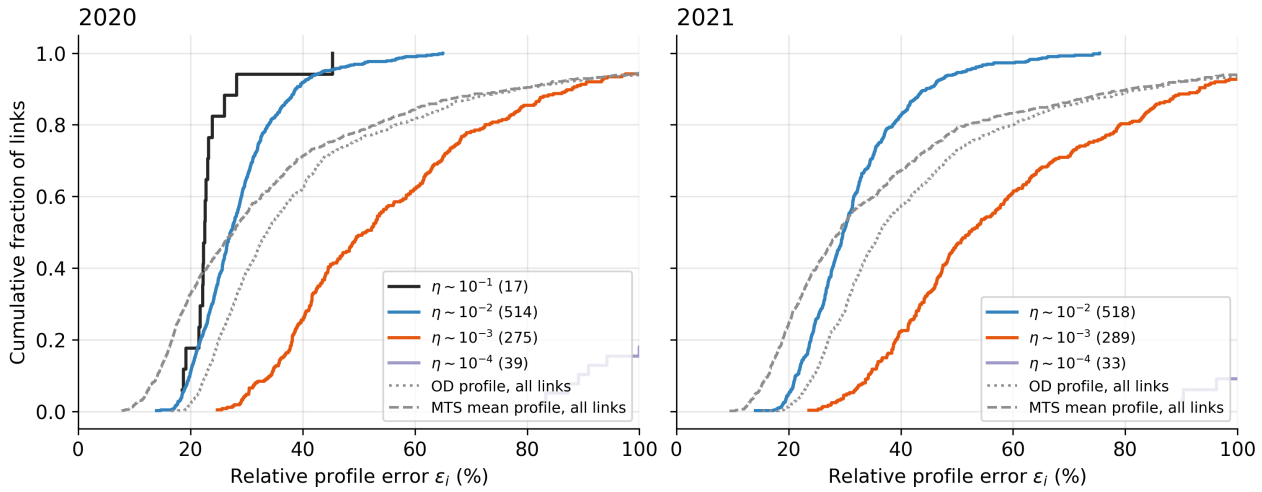


Fig. 8 Profile error on all usable matched links. Cumulative distribution of the relative error ε_i (Eq. (5)) by penetration class, with the number of links in parentheses, for 10–17 February 2020 (left) and 28 June–4 July 2021 (right). Grey curves: the same error when the OD demand profile (dotted) or the MTS mean profile (dashed) is used as prediction on every link.

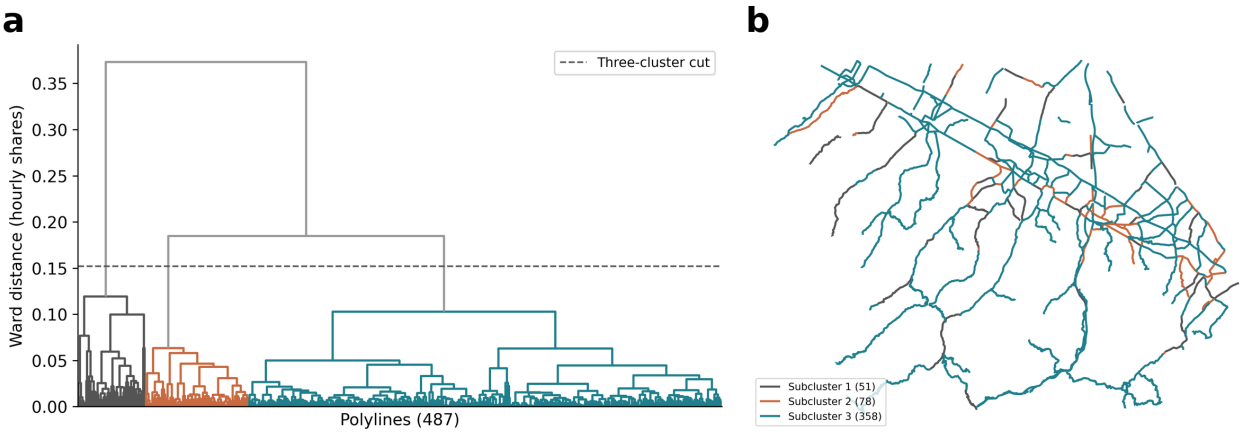


Fig. 9 Subclusters of the high-penetration links in 2020. (a) Ward dendrogram of the normalized hourly profiles of the 487 candidate links; the dashed line marks the three-cluster cut. **(b)** Location of the three subclusters, with the number of links in parentheses.

One link per subcluster is shown in Fig. 10. Polyline 104671 is kept because it lies close to the urban area of Forlimpopoli. The other two are the links of subclusters 2 and 3 with the lowest MAE among those at least 40 km from the other representatives: polylines 134419 and 64544. They are therefore best cases. Their MAE is 0.81 and 0.72 pp ($\varepsilon = 19\%$ and 17%), lower than on 95% and 99% of the usable links, while polyline 104671 has MAE = 1.22 pp ($\varepsilon = 29\%$), close to the median of the $\eta \sim 10^{-2}$ class (Supplementary Section S5). On all three roads the simulated morning rise precedes the GPS one by about an hour, which is the lead discussed above.

2.7 Operational webserver

The same chain runs as a webserver. An authenticated user selects a time interval in the browser; the backend builds the model configuration from the OD tables and the available coil data, runs `minimocas` in a separate process, applies the coil assimilation and stores the link-by-hour table with the run metadata (user, start time, runtime, status). Users can download the assimilated flows indexed by polyline and hour, plot the time series of selected links and map all polylines coloured by flow quantile. The dashboard also shows the MTS data (station maps, time series, rankings, clusters and anomaly diagnostics), so that counter quality and assimilation can be inspected in one place. Supplementary Section S7 describes the components, the user workflow and the API.

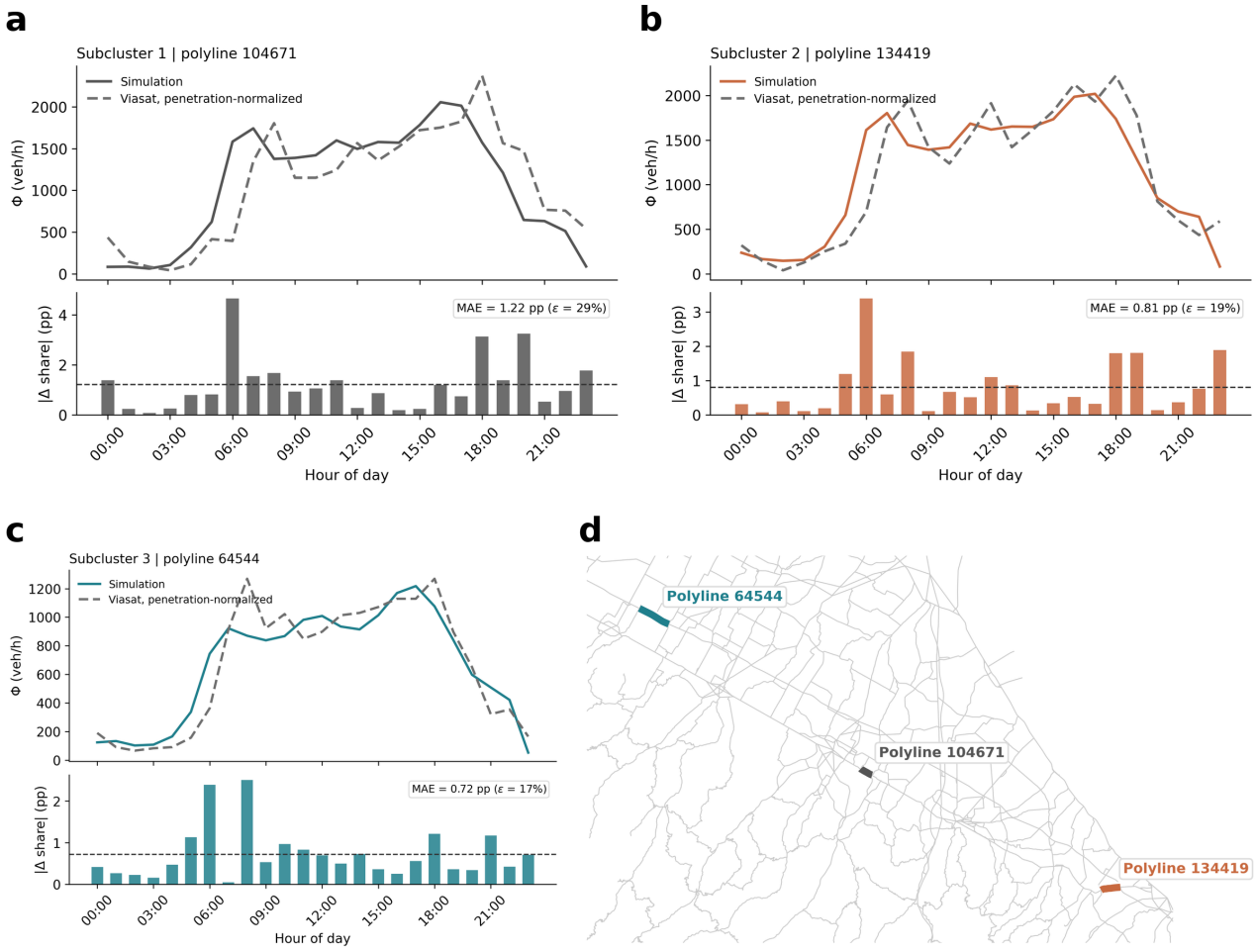


Fig. 10 Representative links, 10–17 February 2020. (a–c) Mean simulated hourly flow and penetration-normalized Viasat profile (top) and hourly absolute share difference $|p_i(h) - q_i(h)|$ (bottom). **(a)** Subcluster 1, polyline 104671, MAE = 1.22 pp. **(b)** Subcluster 2, polyline 134419, MAE = 0.81 pp. **(c)** Subcluster 3, polyline 64544, MAE = 0.72 pp. **(d)** Location of the three polylines on the road graph, coloured as in Fig. 9.

3 Methods

3.1 Data sources and preprocessing

The road support is a routable regional graph derived from OpenStreetMap cartography. Road polylines are represented by local and cartographic identifiers, front and tail nodes, length and direction metadata. A preprocessed cartography file defines the node–polyline adjacency used both by the simulator and by the data-assimilation algorithm. The preprocessing follows standard graph-based principles for converting detailed cartography into a transport network that is simple enough to simulate while preserving the topological relations relevant to routing [35, 36, 43, 44].

Mobile-phone mobility is aggregated to municipality-level OD activity. The January 2020 records are separated into weekdays (Monday–Friday) and weekend days (Saturday–Sunday), with the public holidays of 1 and 6 January counted as weekdays, and into four generation classes: observed source–observed destination (`src-dest`), observed source–random destination (`src-rnd`), random source–observed destination (`rnd-dest`) and random source–random destination (`rnd-rnd`). These classes determine how origins and destinations are sampled when demand is injected into the simulation. Random endpoints, drawn with a travel-time budget, are

used only for residual demand that cannot be assigned to two observed municipalities, because trip-distribution models are sensitive to spatial support and missing observations [24, 25, 32, 33].

MTS coil data are matched to road polylines through barrier metadata. A direction-pivot table maps registry directions to cartographic directions (`IN`/`OUT`). In the archived 2020 runs, short gaps in the observed coil series are filled by linear interpolation in time; for the 2021 runs no observed table in the project archive covers the simulated days, and the code falls back to weekday-specific seasonal average profiles. Viasat GPS data are map-matched to road polylines independently. The meaning, scale and role of each data layer are summarized in Supplementary Section S1.

3.2 The `minimocas` traffic model

`minimocas` is an agent-based traffic simulator in which each vehicle is represented by an agent called `"pawn"`. A pawn has a current polyline, a position along the polyline, a speed, a destination and a status. Simulation time advances with time step Δt; in the Emilia-Romagna configuration the code supports sub-hourly evolution and hourly camera dumps. During each time step, active pawns move along their current polyline. If they reach a node, the next polyline is chosen from the best-path matrix for the current destination, with an optional probability of missing the best path. The best-path matrix is computed on the weighted graph with a Dijkstra-style engine and respects no-turn constraints. Routing is shortest-path without adaptive rerouting, and congestion enters only through the speed rule described below; these choices keep regional runs stable and fast. The model is simpler than full microscopic platforms such as MATSim and SUMO [40, 41], but it shares their agent-based representation of trips and network loading.

Sources inject new pawns according to time-varying creation rates. The daily creation-rate vector is indexed by simulation time, allowing the mobile-phone hourly demand profile to modulate the number of generated trips. Depending on the generation class, the model can use observed origins and destinations, draw one missing endpoint conditionally, or draw both endpoints from the regional node set. The selected OD pair is converted into an initial directed polyline and a destination node; the pawn then follows the network until the trip is completed or killed by model rules.

The simulator records two forms of traffic output. Barrier counters count crossings at coil-like virtual barriers. Unique polyline output records, for each dump interval and each direction, the number of distinct pawn identifiers present on a polyline. This second output is the link-level quantity used in the reconstructed flow maps and in the subsequent data-assimilation step.

The model also updates link velocities from simulated density. For each polyline i, density is the number of pawns on the link divided by its length. The code imposes a critical-density rule: above the critical density the link speed is set to a critical speed; below it, the nominal polyline speed is linearly reduced with density. The link flow used internally is the product of density and updated speed. This places the implementation within the broader family of traffic-flow and car-following models that link microscopic motion to macroscopic density, speed and flow [42, 45–52]. Supplementary Section S2 provides the run inventory and source-class counts, while Supplementary Section S3 describes the model-state variables.

3.3 Coil-based data assimilation

The data-assimilation procedure implemented in the framework adjusts the simulated unique-link matrix using MTS coil residuals. Let $u_i(h)$ denote the simulated unique count on the polyline associated with coil i, $b_i(h)$ the corresponding synthetic barrier count, and $c_i(h)$ the observed coil count after direction mapping and temporal harmonization. The code first estimates the relation between barrier and unique counts,

$$r_i(h) = \frac{b_i(h)}{\max(u_i(h), 1)}, \tag{6}$$

and computes a relative residual in barrier space,

$$\Delta_i(h) = \frac{c_i(h) r_i(h) - b_i(h)}{\max(b_i(h), 1)}. \tag{7}$$

The residual is turned into an additive increment on the directed coil link,

$$\delta_i(h) = u_i(h)\,\Delta_i(h), \qquad \tilde{u}_i(h) = u_i(h) + \delta_i(h), \tag{8}$$

which is applied only when $\tilde{u}_i(h) > 0$. Whenever $u_i(h) \geq 1$ and $b_i(h) \geq 1$, Eqs. (6)–(8) give $\tilde{u}_i(h) = c_i(h)$: on the coil link the assimilated count equals the observation. Negative counts produced by the propagation are clipped to zero. Fixed counters therefore constrain the simulated field where they exist, and the simulation supplies the spatial structure used to pass this information to nearby links [28, 31]; this is a local correction, not a calibration of the whole network.

The increment is propagated recursively through adjacent directed links with turning weights taken from the simulated unique counts. For an incoming link (n_1, n_2) and an admissible outgoing link (n_2, n),

$$\delta_{(n_2,n)}(h) = w_{(n_1,n_2)\to n}(h)\,\delta_{(n_1,n_2)}(h), \qquad w_{(n_1,n_2)\to n}(h) = \frac{u_{(n_2,n)}(h)}{\sum_m u_{(n_2,m)}(h)}, \tag{9}$$

where the sum runs over the admissible outgoing links; when their simulated counts are missing the split is uniform. A backward recursion with the analogous upstream weights distributes the same increment to the incoming links. In the runs analysed here the recursion stops when the propagated increment falls below one vehicle, after three links, when the next node gives access to another coil polyline, or when a forbidden turn or a closed polyline would be crossed. The assimilated table keeps the directional unique-polyline format of the simulator output. Areas far from reliable counters receive no correction and remain as uncertain as the raw simulation. Supplementary Section S4 lists the stopping rules and the differences between the archived implementation and the one now running in the webserver.

3.4 Accident exposure and hazard-weighted traffic

The accident data are the hourly tables of road accidents with injuries in Emilia-Romagna for 2022, 2023 and 2024, averaged over the three years; they pool all days of the week. Exposure comes from twelve weekday runs, one in each month of 2023, and eleven Sunday runs of the same year, all driven by the OD matrices of Section 3.1.

Let $\phi_i(h)$ be the simulated flow on link i during hour h. Hourly exposure is obtained by summing simulated traffic over the road network,

$$E(h) = \frac{\sum_i \phi_i(h)}{\sum_{h'} \sum_i \phi_i(h')}, \tag{10}$$

so that $E(h)$ is a daily share. Accident counts are normalized in the same way,

$$A(h) = \frac{a(h)}{\sum_{h'} a(h')}, \tag{11}$$

where $a(h)$ is the mean number of accidents in hour h, and $R(h)$ follows from Eq. (1). Weekday and weekend exposure give two versions of $R(h)$; the link indicators below use the weekday one.

The hazard-weighted exposure of link i is

$$H_i = \sum_h \phi_i(h) R(h). \tag{12}$$

and the weekday–weekend contrast is

$$D_i = \frac{H_i^{\text{weekday}} - H_i^{\text{weekend}}}{H_i^{\text{weekday}} + H_i^{\text{weekend}}}, \tag{13}$$

evaluated where the denominator is positive. The risk centrality

$$C_i = \frac{H_i}{\sum_j H_j}, \tag{14}$$

is the share of the total hazard-weighted exposure carried by link i, and the rank difference

$$S_i = \text{rank}(H_i) - \text{rank}(E_i), \tag{15}$$

compares the percentile rank of H_i with that of the total simulated exposure $E_i = \sum_h \phi_i(h)$ among the links with $E_i > 0$. Figure 6e highlights the links in the top 5% of S_i.

3.5 GPS validation and clustering

Viasat data enter only after map matching to the simulated polylines, and are not used to generate demand, assimilate coils, tune the simulation or choose parameters [13, 30, 31]. Eqs. (2)–(5) define the penetration, the normalized profile and the profile errors; links with zero simulated or GPS total are excluded. For each

campaign the simulated days are those of the Viasat extract, so the 5 July 2021 run is not used. The baselines of Table 2 are applied with the same formulas, replacing $p_i(h)$ by the baseline profile.

The examples of Fig. 10 are chosen as described in the Results: Ward clustering of the normalized simulated profiles of the high-penetration links, one link per subcluster, with polyline 104671 fixed and the other two selected by lowest MAE under a 40 km separation constraint. Supplementary Section S5 lists the selection steps and the metrics of the selected links.

4 Discussion

The workflow keeps the role of each sensor explicit: mobile-phone records give the regional demand, the road graph and the simulator assign it, coils correct the simulated counts where they exist, and Viasat is kept for validation. Large mobility data sets are most useful when their observation process stays visible [1, 3, 4, 34].

On all usable matched links the median relative profile error is 32% in 2020 and 36% in 2021, and 27% on the links with $\eta \sim 10^{-2}$. The error falls with penetration, as expected for a probe signal. The baselines set these numbers in context: the reconstruction improves only slightly on the timing of the injected demand and is less accurate than the mean profile of the MTS coils, largely because it anticipates the observed peaks by one hour. The leave-one-out test shows that the assimilation fixes the volume on the coil links and changes the error on a held-out coil by about three points. The reconstruction therefore provides a volume for every directed link with a daily shape close to the observed one but no better than a generic counter profile, and its volumes away from coils are not validated by the present data.

The output is a road-level traffic field built from data sources that are usually analysed separately. It can be used to locate corridors with high flows, compare weekday and weekend demand, inspect the profile of specific roads, and provide exposure for emissions, noise or road-safety analyses. Being indexed by polyline and hour, it can be aggregated to municipalities, provinces, corridors or user-defined areas without new runs. Planning tools and regional digital twins need this kind of intermediate layer [2, 5].

The accident analysis uses the reconstructed field as an exposure denominator. With aggregate tables, normalization changes the reading of the hourly counts: the hours around midnight have few accidents but an even smaller simulated exposure, and therefore the highest $R(h)$, while the busy day hours fall below one. The exponent $\alpha = 0.72$ (0.58–0.88) means that across hours the accident share grows less than proportionally with exposure. Speed regime, visibility, road environment and reporting can all contribute [21, 22], and the present data cannot separate them. Because $R(h)$ is regional, the link maps rank roads by the hourly composition of their simulated traffic; geolocated accident records would be needed to test whether the highlighted links are more dangerous.

The webserver turns the offline chain into a repeatable service with the same model and assimilation: a user chooses a time window, runs the simulation, downloads the assimilated flows and maps them, next to the MTS monitoring views (Supplementary Section S7).

4.1 Limitations

The data layers do not cover the same period. The demand is the January 2020 OD matrix in every run. The validation weeks are 10–17 February 2020, a few weeks after the demand period, and 28 June–4 July 2021, a summer week one and a half years later; the accident tables cover 2022–2024 and the exposure runs are days of 2023. All runs therefore use a winter, pre-pandemic demand, including those for summer 2021 and for 2023. The 2021 errors are only slightly larger than the 2020 ones (median MAE 1.49 against 1.34 pp), so the daily shape of the demand seems to change less than its level, but the present data cannot test volumes. Within each validation week the composition of days is the same on both sides: weekend days are simulated with the weekend OD matrix, and the archived 2021 set contained one extra Monday run (5 July), a copy of the 28 June configuration, which we excluded because the Viasat extract ends on 4 July. The accident tables pool all days of the week, while the link indicators use weekday exposure.

The timing of the simulated profiles has two known defects. The simulated peaks precede the Viasat ones by about one hour (Section 2.5). Both series label each hour by its start, and we could not identify the cause from the archived inputs; candidates are the time convention of the hourly OD aggregation and the injection of all trips at their origin without a spread of departure times. In addition, the configuration reads only 23 of the 24 hourly OD values, so no trips are generated between 23:00 and 24:00 and every simulated profile drops in the last hour of the day. Both defects weigh most on the night hours, where $R(h)$ reaches its largest values: at 00:00 the simulated share of daily traffic on the usable Viasat links is 0.65% against 1.05% for Viasat in 2020.

The MTS tables store UTC timestamps, and the assimilation code compares them with the local clock of the simulation without conversion, so in February 2020 each coil hour is imposed one hour early. Rerunning the 2020 assimilation with the coil timestamps converted to local time lowers the median MAE on the Viasat links from 1.34 to 1.31 pp; the figures and tables of this paper use the archived assimilation.

The simulator is deliberately simple, so that regional runs stay compatible with webserver use. It has no dynamic congestion mechanism and no adaptive assignment: trips follow shortest paths with a free-flow speed reduced above a density threshold. This is adequate for aggregate directional flows on a large extra-urban network and inadequate for urban congestion waves. A demand basis with several OD matrices, covering different seasons and possibly different sources, would allow seasonal changes to be reproduced explicitly and the reconstruction to be tested under different demand regimes.

Future work should first correct the timing defects and the coil clock, then extend the demand basis. On the operational side, the webserver could be extended with explicit scenario controls, seasonal demand selection, user-defined regions of interest and automatic reporting of the OD and sensor layers used in each run.

5 Conclusions

We reconstructed hourly traffic flows on the regional road network of Emilia-Romagna from mobile-phone OD activity, inductive loop detectors, OpenStreetMap cartography and Viasat GPS telemetry, keeping demand generation, network loading, local assimilation and validation as separate steps.

Against Viasat, the reconstructed hourly profiles have a median relative error of 32% (2020) and 36% (2021) on all usable links, and 27% where the probe penetration is near 1%. The simulation leads the observed peaks by one hour, gains little over the timing of the demand itself and is less accurate than a mean coil profile; the coil assimilation corrects the instrumented links and has a small effect on the others. These results point to the parts of the chain to improve first: the time convention of the demand, the clock of the coil data and the reach of the assimilation.

The same field can serve as an exposure layer for aggregate accident statistics. Normalized by simulated exposure, the hourly accident share of 2022–2024 gives a relative risk that peaks around midnight and stays below one after 19:00, and across hours the accident share scales as $E^{0.72}$. The link maps derived from this regional profile rank roads by the hourly composition of their traffic, and can indicate where geolocated accident data or local counts should be collected first.

The webserver delivers the assimilated link flows to regional analysts who do not run the modelling chain. Further work should add uncertainty estimates for the link volumes, a demand basis covering several seasons, and validation on other territories with geolocated accident data and denser counter networks.

Supplementary Material

The Supplementary Material documents the data layers, simulation runs, `minimocas` state variables, assimilation rules and leave-one-out test, validation tables, accident descriptors and webserver architecture, and gives reproducibility notes.

Availability of data and materials

The manuscript sources, the analysis scripts and the aggregate result tables are available at https://github.com/federicobellisardi/multi-sensor-traffic-reconstruction. The `minimocas` simulator is available at https://github.com/physycom/minimocas. **[TODO: check that both repositories are public before submission]** Public traffic-counter data from the Emilia-Romagna Region are available through the MTS online mobility platform at https://mobilita.regione.emilia-romagna.it/servizi/flussi-online. Road-accident statistics are publicly available from ISTAT at https://www.istat.it/tag/incidenti-stradali/. Mobile-phone and Viasat GPS datasets used in this study are proprietary and cannot be publicly distributed.

Ethics approval and consent to participate

The mobile-phone data were provided by the operator already anonymized and aggregated, as hourly counts of devices moving between census areas (ACE), which we further aggregated to municipalities. No individual-level data were accessed or processed, and the data were handled in compliance with the EU General Data Protection Regulation (GDPR). **[TODO: confirm with the data provider the aggregation level of the delivered data (ACE or municipality)]** Consent to participate was not applicable.

Competing interests

C.M. and A.F. are employed by Mindicity, which develops the operational platform described in this work. The other authors declare no competing interests.

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Authors' contributions

FB conceived part of the original study design, developed the reconstruction workflow, performed the analyses and prepared the first manuscript draft. CM and AF contributed to the development of the modelling framework. AB contributed to the study design, interpretation of the modelling results and manuscript revision. All authors read and approved the final manuscript.

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